

# Northeast Publication Tables

Table 1: Operating Overrides and Moody's Ratings

|                                 | Moody's credit rating |                      |                      |
|---------------------------------|-----------------------|----------------------|----------------------|
|                                 | (1)                   | (2)                  | (3)                  |
| Override attempt                | 0.130***<br>(0.048)   |                      |                      |
| Successful override             |                       | 0.187***<br>(0.063)  |                      |
| Failed override                 |                       |                      | 0.026<br>(0.055)     |
| Population size (log)           | -1.447***<br>(0.140)  | -1.550***<br>(0.138) | -1.475***<br>(0.139) |
| Outstanding debt level (%)      | -0.003***<br>(0.001)  | -0.003**<br>(0.001)  | -0.003**<br>(0.001)  |
| Unemployment rate (%)           | 0.125***<br>(0.038)   | 0.126***<br>(0.039)  | 0.127***<br>(0.038)  |
| Revenue stability (%)           | -0.009<br>(0.008)     | -0.010<br>(0.008)    | -0.008<br>(0.008)    |
| Government revenue (per capita) | -0.294<br>(0.179)     | -0.293<br>(0.178)    | -0.323*<br>(0.182)   |
| Excess property tax capacity    | -0.867<br>(0.579)     | -0.875<br>(0.579)    | -0.840<br>(0.585)    |
| Fiscal reserve (%)              | 0.050***<br>(0.009)   | 0.050***<br>(0.009)  | 0.049***<br>(0.009)  |
| Budget balance (%)              | -0.010*<br>(0.005)    | -0.010*<br>(0.005)   | -0.010*<br>(0.005)   |
| Observations                    | 3,550                 | 3,550                | 3,550                |
| Municipalities                  | 265                   | 265                  | 265                  |
| Mundlak means                   | Yes                   | Yes                  | Yes                  |
| Year indicators                 | Yes                   | Yes                  | Yes                  |

*Notes:* The dependent variable is Moody's ordered municipal credit rating. Models are ordered probit regressions. Ordered-probit cutpoints are omitted. All columns include the listed controls. Standard errors in parentheses are clustered by municipality. Moody's complete-case years are 2003–2015, 2017, and 2019–2021. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 2: Override Frequency and Moody's Ratings

|                                 | Moody's credit rating |                      |                      |
|---------------------------------|-----------------------|----------------------|----------------------|
|                                 | (1)                   | (2)                  | (3)                  |
| 3-year override attempts        | -0.009<br>(0.025)     |                      |                      |
| 3-year successful overrides     |                       | -0.014<br>(0.051)    |                      |
| 3-year failed overrides         |                       |                      | -0.010<br>(0.022)    |
| Population size (log)           | -1.539***<br>(0.139)  | -1.590***<br>(0.137) | -1.547***<br>(0.139) |
| Outstanding debt level (%)      | -0.003***<br>(0.001)  | -0.003***<br>(0.001) | -0.003**<br>(0.001)  |
| Unemployment rate (%)           | 0.129***<br>(0.039)   | 0.128***<br>(0.038)  | 0.128***<br>(0.038)  |
| Revenue stability (%)           | -0.008<br>(0.008)     | -0.008<br>(0.008)    | -0.008<br>(0.008)    |
| Government revenue (per capita) | -0.325*<br>(0.182)    | -0.324*<br>(0.182)   | -0.325*<br>(0.181)   |
| Excess property tax capacity    | -0.865<br>(0.580)     | -0.876<br>(0.591)    | -0.840<br>(0.582)    |
| Fiscal reserve (%)              | 0.049***<br>(0.009)   | 0.049***<br>(0.009)  | 0.049***<br>(0.009)  |
| Budget balance (%)              | -0.009*<br>(0.005)    | -0.009*<br>(0.005)   | -0.009*<br>(0.005)   |
| Observations                    | 3,550                 | 3,550                | 3,550                |
| Municipalities                  | 265                   | 265                  | 265                  |
| Mundlak means                   | Yes                   | Yes                  | Yes                  |
| Year indicators                 | Yes                   | Yes                  | Yes                  |

*Notes:* The dependent variable is Moody's ordered municipal credit rating. Models are ordered probit regressions. Ordered-probit cutpoints are omitted. All columns include the listed controls. Standard errors in parentheses are clustered by municipality. Moody's complete-case years are 2003–2015, 2017, and 2019–2021. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 3: Annual Override Frequency and Yes-Vote Percentage

|                                 | Yes-vote percentage  |                      |                      |
|---------------------------------|----------------------|----------------------|----------------------|
|                                 | (1)                  | (2)                  | (3)                  |
| Annual override attempts        | -0.657***<br>(0.189) |                      |                      |
| Annual successful overrides     |                      | 2.618***<br>(0.659)  |                      |
| Annual failed overrides         |                      |                      | -2.025***<br>(0.351) |
| Population size (log)           | 19.314<br>(21.973)   | 41.588*<br>(24.104)  | 18.287<br>(21.260)   |
| Outstanding debt level (%)      | -0.065***<br>(0.020) | -0.045**<br>(0.018)  | -0.065***<br>(0.020) |
| Unemployment rate (%)           | 1.350**<br>(0.532)   | 1.305**<br>(0.521)   | 1.373***<br>(0.477)  |
| Revenue stability (%)           | 0.534***<br>(0.175)  | 0.439***<br>(0.148)  | 0.470**<br>(0.197)   |
| Government revenue (per capita) | 4.166***<br>(1.468)  | 4.935***<br>(1.517)  | 3.973***<br>(1.330)  |
| Excess property tax capacity    | 8.272<br>(9.269)     | 2.358<br>(10.107)    | 4.509<br>(9.365)     |
| Fiscal reserve (%)              | 0.075<br>(0.116)     | 0.110<br>(0.116)     | 0.096<br>(0.116)     |
| Budget balance (%)              | -0.242**<br>(0.096)  | -0.285***<br>(0.089) | -0.159<br>(0.099)    |
| Observations                    | 730                  | 730                  | 730                  |
| Municipalities                  | 221                  | 221                  | 221                  |
| Municipality fixed effects      | Yes                  | Yes                  | Yes                  |
| Year indicators                 | Yes                  | Yes                  | Yes                  |

*Notes:* The dependent variable is operating override yes-vote percentage. Models are linear fixed-effects regressions. All columns include the listed controls. Standard errors in parentheses are clustered by municipality. Complete-case years are 2003–2021. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 4: Three-Year Override Frequency and Yes-Vote Percentage

|                                 | Yes-vote percentage  |                      |                      |
|---------------------------------|----------------------|----------------------|----------------------|
|                                 | (1)                  | (2)                  | (3)                  |
| 3-year override attempts        | -0.156<br>(0.162)    |                      |                      |
| 3-year successful overrides     |                      | 1.066***<br>(0.239)  |                      |
| 3-year failed overrides         |                      |                      | -0.875***<br>(0.241) |
| Population size (log)           | 23.686<br>(21.331)   | 31.731<br>(24.228)   | 19.320<br>(21.273)   |
| Outstanding debt level (%)      | -0.061***<br>(0.020) | -0.053***<br>(0.019) | -0.062***<br>(0.020) |
| Unemployment rate (%)           | 1.394**<br>(0.549)   | 1.122**<br>(0.543)   | 1.520***<br>(0.532)  |
| Revenue stability (%)           | 0.552***<br>(0.164)  | 0.410***<br>(0.158)  | 0.550***<br>(0.169)  |
| Government revenue (per capita) | 4.340***<br>(1.512)  | 4.537***<br>(1.478)  | 3.948***<br>(1.341)  |
| Excess property tax capacity    | 7.518<br>(9.437)     | 5.447<br>(8.862)     | 3.258<br>(9.179)     |
| Fiscal reserve (%)              | 0.071<br>(0.118)     | 0.107<br>(0.118)     | 0.066<br>(0.113)     |
| Budget balance (%)              | -0.267***<br>(0.093) | -0.304***<br>(0.091) | -0.212**<br>(0.094)  |
| Observations                    | 730                  | 730                  | 730                  |
| Municipalities                  | 221                  | 221                  | 221                  |
| Municipality fixed effects      | Yes                  | Yes                  | Yes                  |
| Year indicators                 | Yes                  | Yes                  | Yes                  |

*Notes:* The dependent variable is operating override yes-vote percentage. Models are linear fixed-effects regressions. All columns include the listed controls. Standard errors in parentheses are clustered by municipality. Complete-case years are 2003–2021. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 5: Repeated-Event DiD Estimates for Rating Downgrades

|                                     | Rating downgrade  |                      |                     |
|-------------------------------------|-------------------|----------------------|---------------------|
|                                     | Attempt           | Success              | Failure             |
| $h = -2$                            | -0.020<br>(0.016) | -0.035***<br>(0.010) | -0.004<br>(0.016)   |
| $h = 0$                             | 0.025<br>(0.022)  | 0.029<br>(0.023)     | 0.018<br>(0.022)    |
| $h = 1$                             | -0.011<br>(0.013) | 0.019<br>(0.020)     | -0.024**<br>(0.012) |
| $h = 2$                             | -0.020<br>(0.015) | 0.001<br>(0.021)     | -0.029**<br>(0.014) |
| Observations                        | 112,960           | 118,666              | 106,167             |
| Municipalities                      | 246               | 257                  | 259                 |
| Stacks                              | 179               | 157                  | 131                 |
| Stack-by-municipality fixed effects | Yes               | Yes                  | Yes                 |
| Stack-by-year fixed effects         | Yes               | Yes                  | Yes                 |
| Covariates                          | No                | No                   | No                  |

*Notes:* Models are repeated-event linear probability models using the window-clean preferred specification. The omitted reference period is  $h = -1$ . The event window is  $h = -2$  to  $h = 2$ . The comparison pool is window-clean controls. Models include stack-by-municipality and stack-by-year fixed effects, with no additional covariates. Standard errors in parentheses are clustered by municipality. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 6: Repeated-Event DiD Estimates for Rating Upgrades

|                                     | Rating upgrade      |                     |                   |
|-------------------------------------|---------------------|---------------------|-------------------|
|                                     | Attempt             | Success             | Failure           |
| $h = -2$                            | -0.017<br>(0.011)   | -0.016<br>(0.010)   | -0.009<br>(0.013) |
| $h = 0$                             | -0.029**<br>(0.014) | -0.029**<br>(0.015) | -0.012<br>(0.020) |
| $h = 1$                             | -0.027<br>(0.019)   | -0.036<br>(0.022)   | -0.015<br>(0.029) |
| $h = 2$                             | -0.013<br>(0.026)   | -0.031<br>(0.026)   | -0.030<br>(0.037) |
| Observations                        | 112,960             | 118,666             | 106,167           |
| Municipalities                      | 246                 | 257                 | 259               |
| Stacks                              | 179                 | 157                 | 131               |
| Stack-by-municipality fixed effects | Yes                 | Yes                 | Yes               |
| Stack-by-year fixed effects         | Yes                 | Yes                 | Yes               |
| Covariates                          | No                  | No                  | No                |

*Notes:* Models are repeated-event linear probability models using the window-clean preferred specification. The omitted reference period is  $h = -1$ . The event window is  $h = -2$  to  $h = 2$ . The comparison pool is window-clean controls. Models include stack-by-municipality and stack-by-year fixed effects, with no additional covariates. Standard errors in parentheses are clustered by municipality. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .